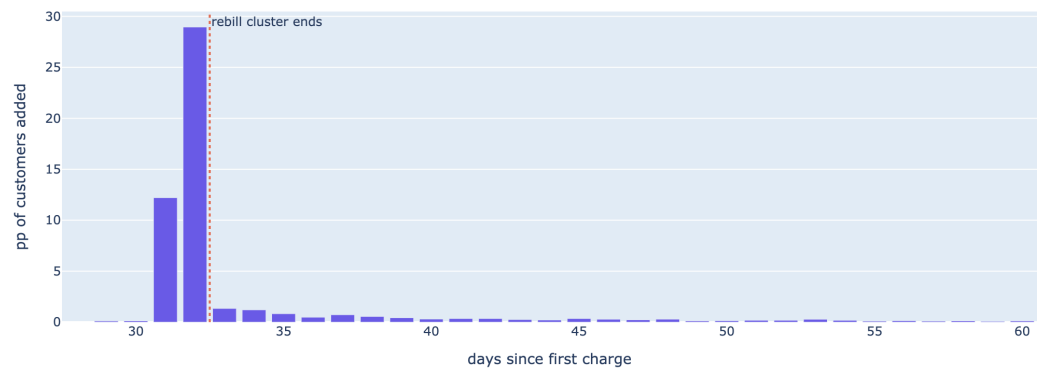


LTV Modeling

1. Rules on data

fact	evidence	consequence
rebills land at 30–32 days	those three days add 41.3pp of renewals; days 33–37 add 4.6pp and days 38–60 a further 5.3pp, at a flat trickle with no second cluster	30-day cycle, grid shifted 2 days so short calendar months land right
no grace period is needed	there is no gap in the distribution for one to sit in — the cycle boundary is already the deadline	dropped
win-backs are a separate event	1,488 Reactivation charges across 1,321 accounts, at a median of 64 days , 4.7% of subscription revenue	recovered revenue, not retention — worth 3.3pp on the first renewal
the last month has no new subscriptions	September: 0 creates, 6,647 rebills	pure retention month — used in full for renewal, excluded from cohort mix

when the second charge arrives: the rebill cluster ends at day 32



Second charges pile up on days 31–32 and then flatten. There is no dip after the cluster for a grace period to sit in, so the cycle boundary is the deadline.

A grace period is a guess at what a late charge means. The data labels it instead.

Annual does not renew monthly

- **87.5%** of the 2,065 annual accounts have exactly **one** subscription charge
- first charge median **\$302.60** vs **\$14.00** monthly — a genuine yearly prepay
- only **3.1%** ever rebill, at a median of **92 days**, for a median **\$42.89** — a downgrade to monthly, not an annual renewal

→ annual renewal is **unmeasurable** here. Excluded from churn and LTV, kept in the mix analysis. A blended churn number would credit annual accounts with retention nobody has observed.

Base

19,530 paying accounts; **19,125** observable (tenure ≥ 32 days), of which **17,138** monthly. Segments: 5,743 / 6,887 / 3,669 / 2,323 / 503

2. Churn

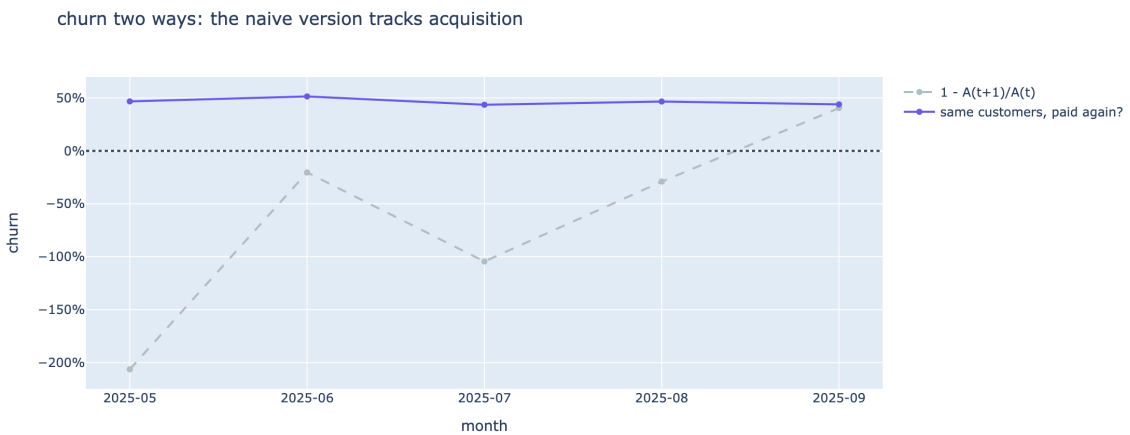
Churned at cycle *m* = no scheduled subscription charge in cycle *m*, measured only on accounts for which cycle *m* has fully elapsed. A lifetime ends at the first missed cycle.

Safe to be this strict: only **2.6%** of accounts miss a cycle and pay again later, so the rule is not quietly counting dunning gaps as churn.

Reported as **chained renewal rates** — each cycle's rate computed on accounts that reached it and paid the cycle before, then multiplied into a survival curve. Accounts too new to have faced a renewal leave the denominator instead of counting as retained. That is the censoring problem handled directly, with no survival model in between.

What it is not

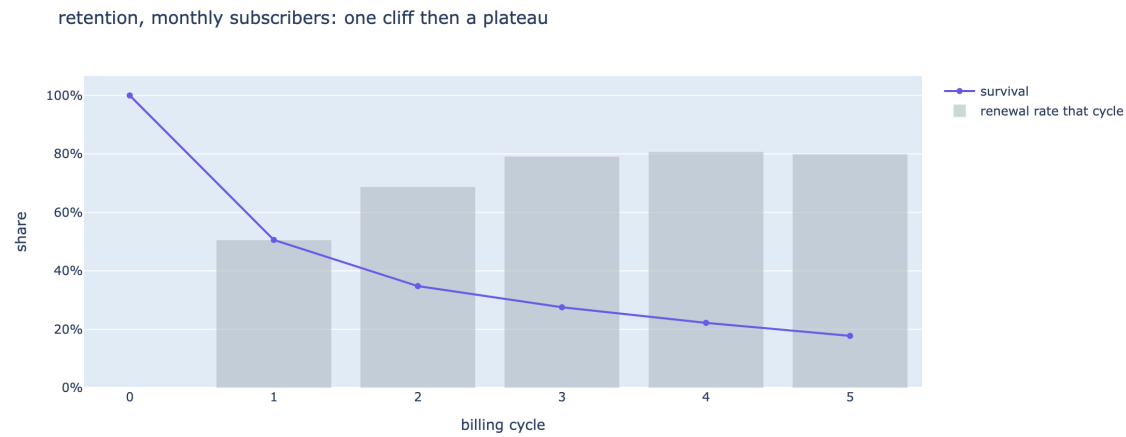
	why not
usage churn	they stop generating before they stop paying — earlier signal, different question
$1 - A(t+1)/A(t)$	negative in four of five months while real churn sat at 43–51%. It tracks acquisition. The one month it looks sane is the month with zero acquisition
voluntary vs involuntary	cannot separate a cancellation from failure



The dashboard formula sits below zero in four of five months — it is measuring acquisition. The cohort version is flat at 43–51%.

Result

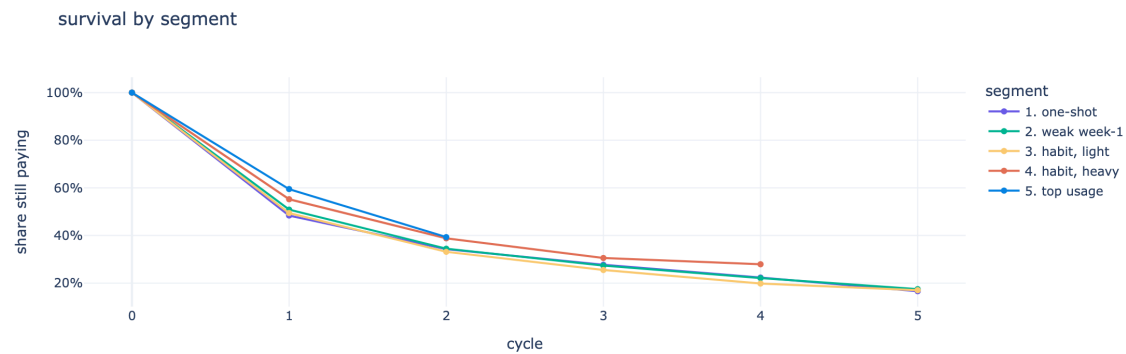
cycle	1	2	3	4	5
renewal	51%	69%	79%	81%	80%
still paying	51%	35%	28%	22%	18%



Bars are the renewal rate of each cycle, the line is the chained survival. The first bar is the only one below 69%.

One cliff, then a plateau. 49% churn at the first bill, 19–31% after it. Anything that moves the first renewal is worth several times the same effort spent later.

Survival at the first renewal by segment: **0.48 / 0.51 / 0.49 / 0.55 / 0.59.**



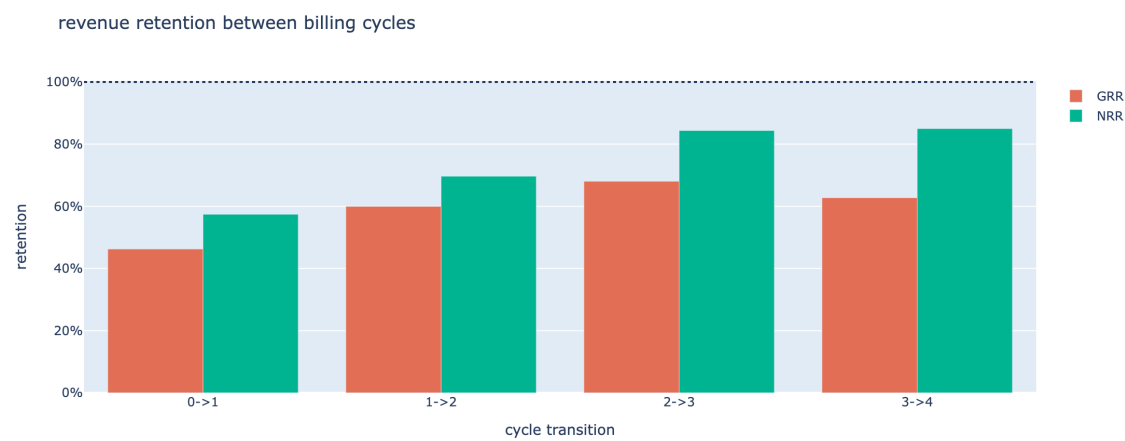
Behaviour reorders the curves but does not change their shape — every segment loses about half of its accounts at the first bill.

3. What churn costs

$$\text{GRR} = \frac{B - \text{churned} - \text{contraction}}{B}, \quad \text{NRR} = \text{GRR} + \frac{\text{expansion}}{B}$$

On billing cycles, not calendar months, so renewals do not smear across boundaries. Credits count as expansion.

transition	0→1	1→2	2→3	3→4
GRR	46%	60%	68%	63%
NRR	57%	70%	84%	85%

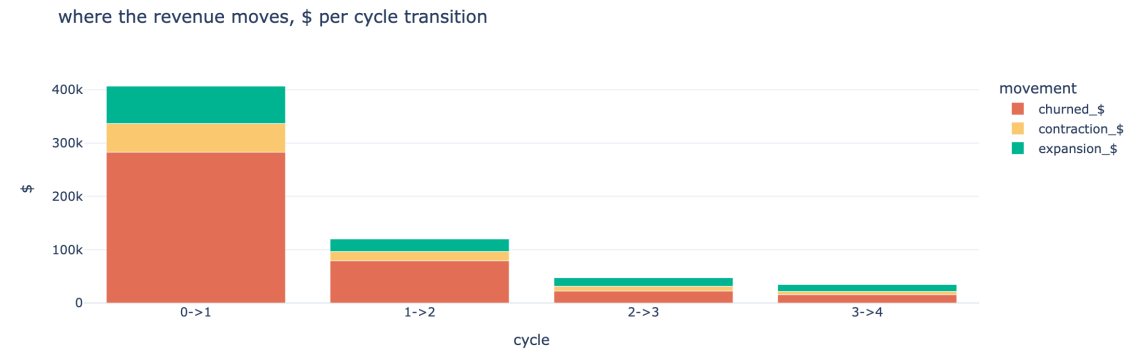


The dotted line is 100%. Expansion lifts every transition, and no bar reaches it.

Survivors expand, but NRR never crosses 100% once win-backs are excluded — **expansion softens churn here, it does not out-run it.**

The leak is proportional

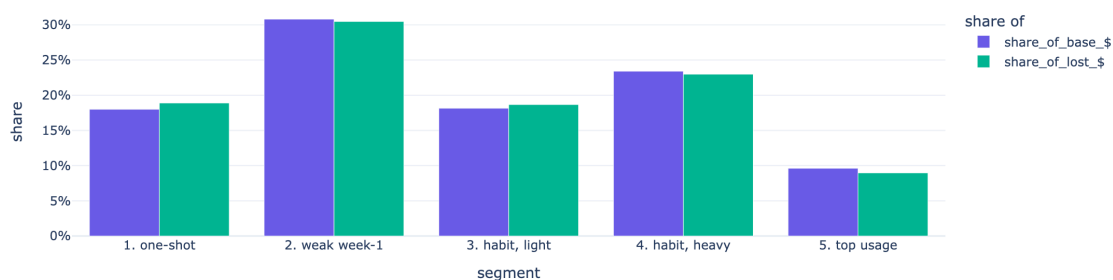
\$302,635 — 48% of first-cycle revenue — never renews.



Dollars per cycle transition. The first transition dominates — churn there is larger than everything that moves in all later cycles combined.

segment	churn	share of cycle-0 \$	share of lost \$
1. one-shot	52%	18%	19%
2. weak week-1	49%	31%	30%
3. habit, light	51%	18%	19%
4. habit, heavy	45%	23%	23%
5. top usage	41%	10%	9%

share of cycle-0 revenue vs share of revenue lost at the first renewal



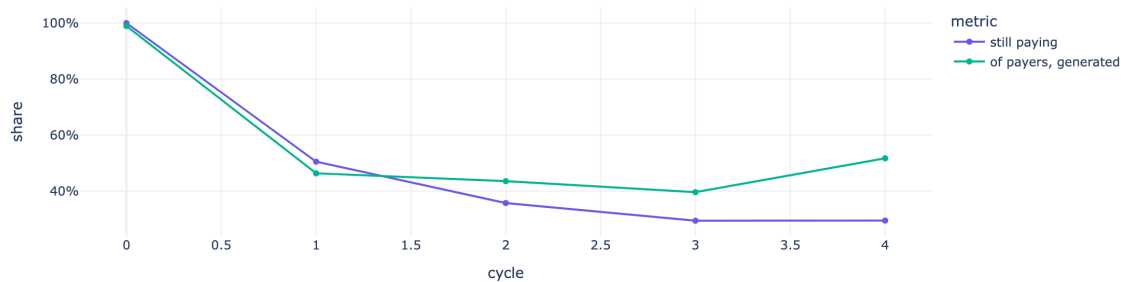
Each segment's share of the money at risk against its share of the money lost. The pairs match, so no segment is the leak.

Churn spans only 52% → 41% across the whole segmentation and every segment loses roughly what it contributes

A third of the revenue is dormant

Of the accounts that renewed once, only **46%** generated anything in the cycle they had just paid for (99% did in cycle 0). Those dormant payers are **\$89,121, 36% of cycle-1 revenue**, and the share stays near a third through cycle 3.

of the accounts still paying, how many still generate



The two lines separate immediately after the first bill: the gap between them is revenue from accounts that are not using the product.

- an **early-warning list available now** — usage stops before payment does
- a **liability** — revenue from people who forgot they subscribed returns as refunds and bad reviews, while sitting inside GRR looking like healthy retention

4. Gross LTV

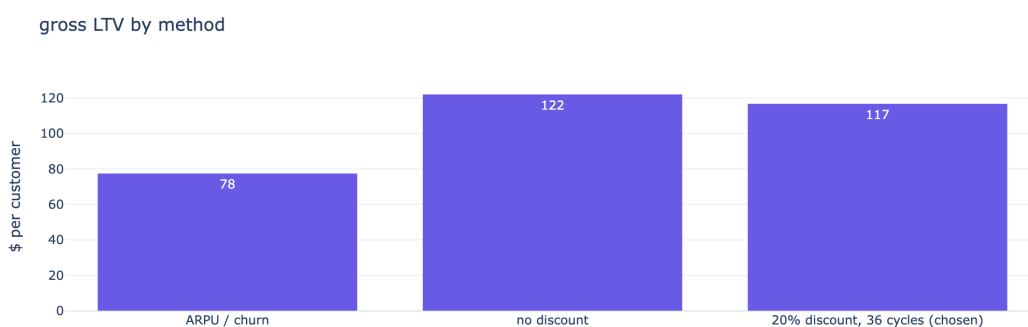
$$\text{LTV}_{\text{gross}} = \sum_{m=0}^{36} \frac{r_m \cdot S(m)}{(1+d)^m}$$

- r_m — observed revenue per surviving customer in cycle m , subscription + credits, so expansion is inside LTV
- $S(m)$ — the chained renewal curve above. No survival model: rates read off the data, at-risk denominator does the censoring work
- d — 20% a year, monthly; horizon capped at 36 cycles

The **extrapolation rule decides the answer**, because five cycles cannot speak for thirty-six:

- a cycle is read off the data only while it still holds 5% of the group
- past that, revenue is held flat and survival decays at the **mean renewal rate of cycles 2+**, excluding cycle 1 as a different regime

method	LTV	what it gets wrong
ARPU / churn	\$78	one churn rate for everybody, from the worst cycle, forever, undiscounted
retention × revenue, flat tail	\$312	assumes the last observed cycle repeats — survival never decays
retention × revenue, decaying tail, discounted	\$117	chosen

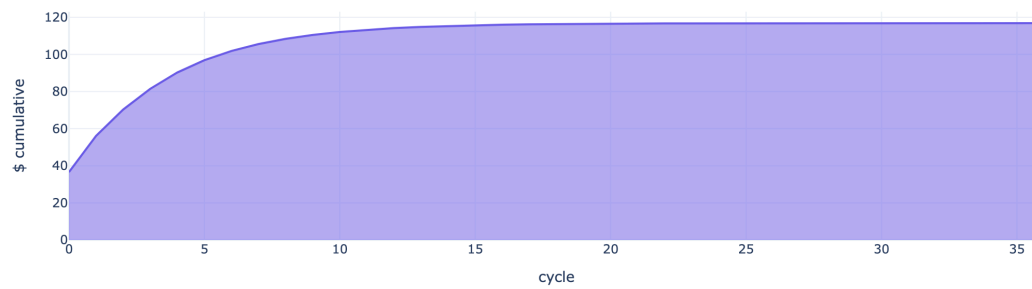


ARPU / churn lands a third low; removing the discount adds \$5. The flat-tail variant is off this scale at \$312.

Two easy mistakes bracket the answer by 4×, and each is one line of code.

98% of LTV lands within 12 cycles → a 12-month payback rule loses almost nothing, and it is the only horizon worth promising on six months of data.

where gross LTV accumulates (present value)

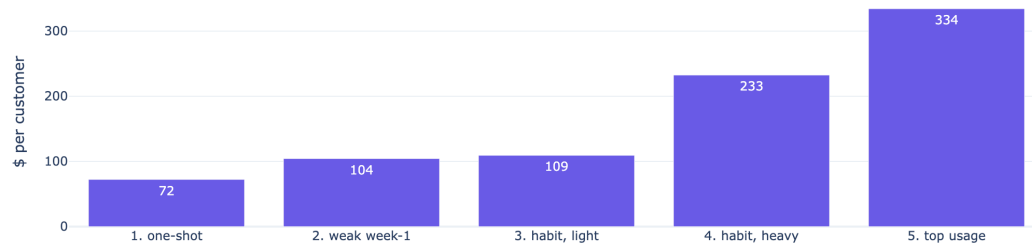


Present value by cycle: the curve is flat well before cycle 36, so the horizon is not what the number rests on.

By segment

	1. one-shot	2. weak week-1	3. habit, light	4. habit, heavy	5. top usage
cycle-0 revenue	\$21.91	\$31.41	\$34.40	\$70.60	\$132.87
survives first renewal	0.48	0.51	0.49	0.55	0.59
gross LTV	\$72	\$104	\$109	\$233	\$334

gross LTV by segment



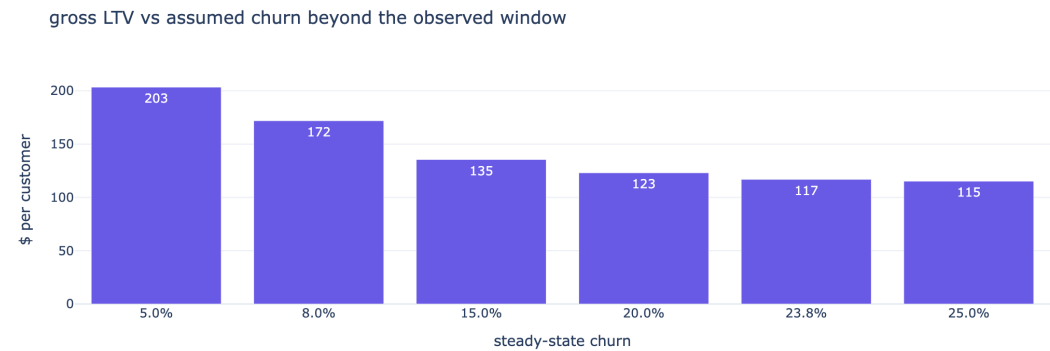
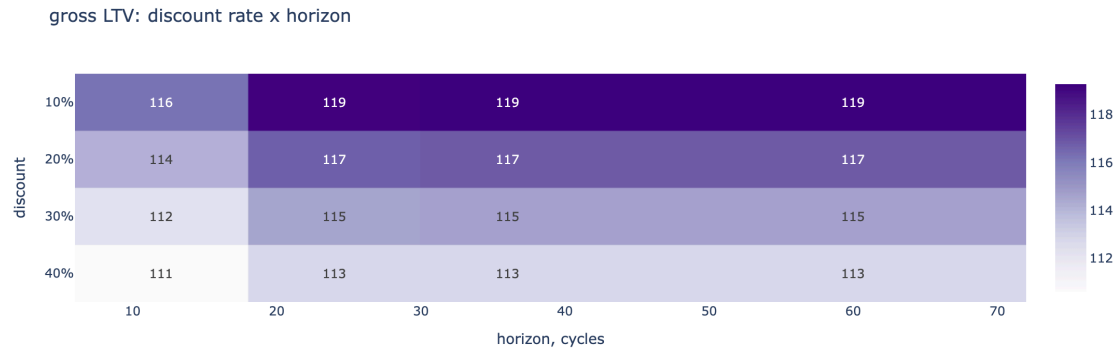
The spread comes from cycle-0 revenue: survival differs by 11pp across segments, revenue by 6x.

A **4.6x spread**, driven by revenue more than by retention → a single blended CAC target overpays for one-shot and underbids for top usage.

What moves the number

assumption	range	factor
discount rate × horizon	\$111–119	1.1×

post-window churn	\$115–203	1.8×
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Discount rate and horizon move the answer by \$8 in total; the assumed steady-state churn moves it by \$88.

→ quote LTV as "**\$117, \$115–203 depending on where churn settles**". The argument to have is about the tail, not the discount rate.

5. Net LTV: the cost ceiling

$$\text{LTV}_{\text{net}} = \sum_m \frac{r_m(1-f) - c_{\text{img}}g^{\text{img}}_m - c_{\text{vid}}g^{\text{vid}}_m}{(1+d)^m} - \text{CAC}$$

Fees are knowable (~3.5%) and **the job counts are in the data**, but cost per job and CAC are not — so any net LTV figure would be an assumption dressed as a result.

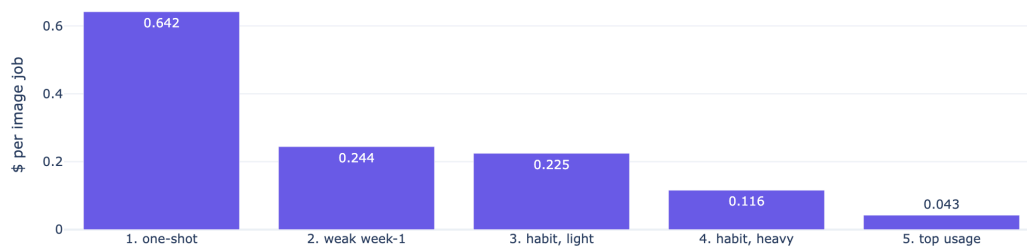
One thing can still be computed exactly: **the compute cost at which a segment stops paying for itself.**

$$c^*_{\text{img}} = \frac{\sum_m r_m(1-f)S(m)/(1+d)^m}{\sum_m \left(g^{\text{img}}_m + k, g^{\text{vid}}_m \right) S(m)/(1+d)^m}, \quad k = c_{\text{vid}}/c_{\text{img}} = 8$$

segment	gross LTV	cost-weighted jobs	breakeven \$/image job
1. one-shot	\$72	109	\$0.64
2. weak week-1	\$104	412	\$0.24

3. habit, light	\$109	470	\$0.22
4. habit, heavy	\$233	1,934	\$0.12
5. top usage	\$334	7,582	\$0.04

breakeven cost per image job (video = 8x), before CAC



The ceiling falls as usage rises: one-shot pays for itself at \$0.64 per image, top usage only below \$0.04.

A 16x spread, in the reverse order of LTV. Top usage carries ~70x the job load of one-shot for 4.6x the revenue.

If cost per image is near \$0.01, top usage is the best customer Higgsfield has. Above ~\$0.04 it is sold at a loss while one-shot stays profitable.

Two missing numbers become one question finance can answer in a minute. Asked in order: **cost per generation**, the video/image cost ratio (assumed 8x), then CAC by channel.

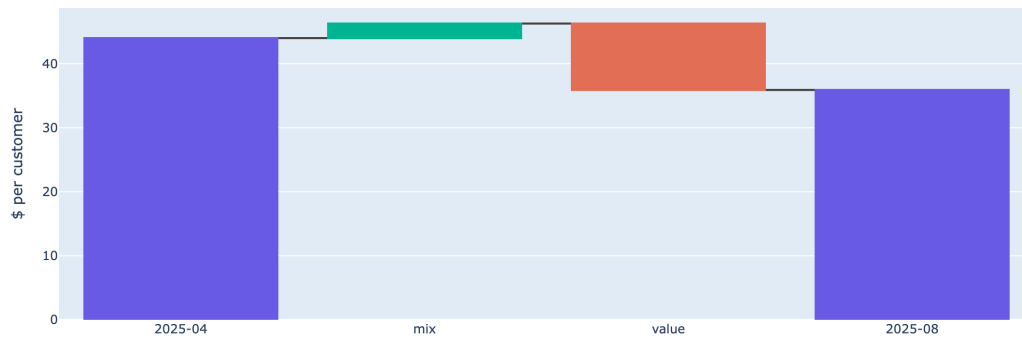
6. How the segments moved

Aggregate value falling has two causes with opposite fixes — **mix shift** (acquisition brings different people) or **within-segment decay** (the same people are worth less). Decomposed symmetrically, so there is no interaction term to argue about:

$$\Delta = \underbrace{\sum_s (w_s^1 - w_s^0) \frac{v_s^0 + v_s^1}{2}}_{\text{mix}} + \underbrace{\sum_s \frac{w_s^0 + w_s^1}{2} (v_s^1 - v_s^0)}_{\text{value}}$$

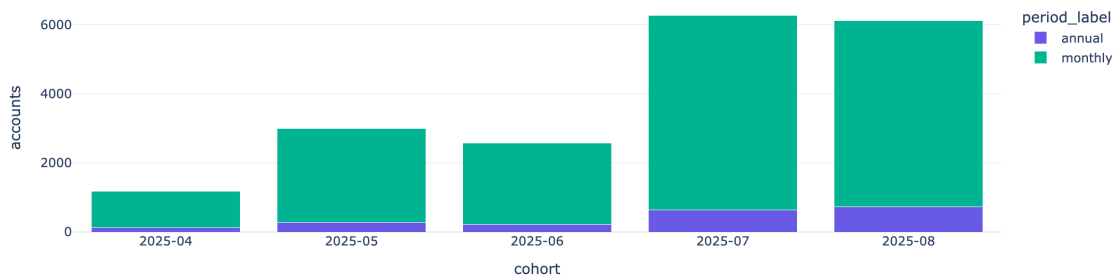
window	ΔARPU	mix	value
Apr → Aug	−\$8.09	+\$2.28	−\$10.36
May → Aug	+\$3.72	+\$7.15	−\$3.43

ARPU change 2025-04 -> 2025-08: mix vs value



Apr → Aug: mix adds \$2.28, within-segment value takes \$10.36 away. Nothing is left over for an interaction term to hide in.

cohort size and billing mix



Why the baseline matters: April is a launch month, and each later cohort is several times its size.

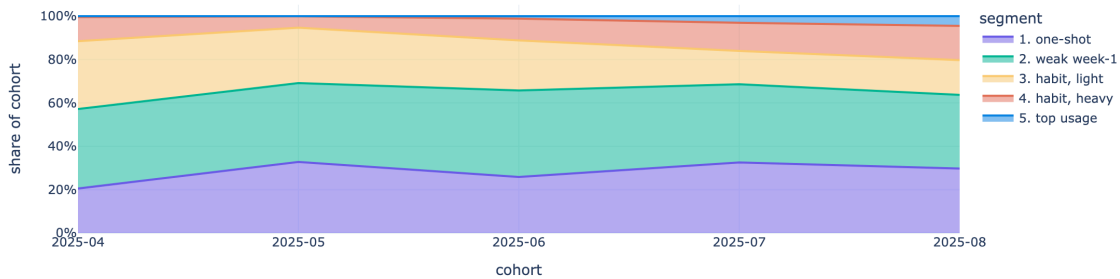
The sign flips with the baseline. April is the launch month — 1,181 accounts, 131 of them heavy — so anchoring a trend on it manufactures a decline.

Honest reading: ARPU has been roughly flat since May.

What both windows agree on

segment	share	30-day ARPU
4. habit, heavy	5% → 17%	\$121 → \$70
3. habit, light	26% → 17%	\$36 → \$32
5. top usage	~0% → 4%	stable ~\$131–140
1. one-shot	33% → 31%	stable ~\$21

segment mix by cohort



ARPU (monthly-equivalent, first 30 days) by cohort

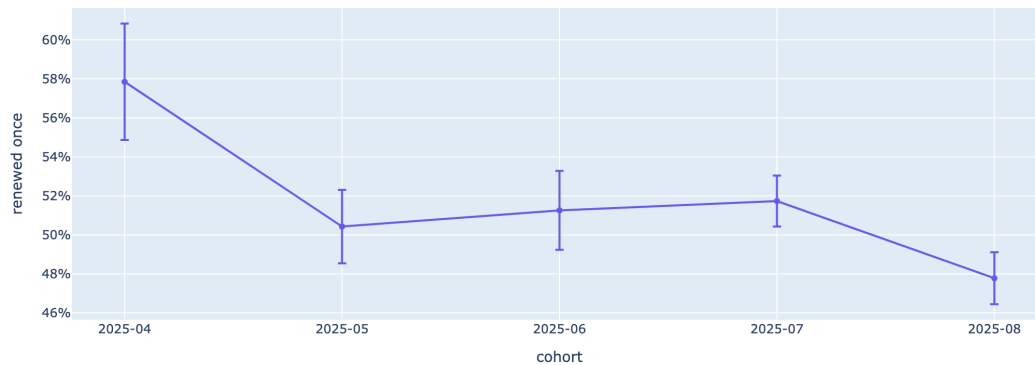


Mix: the heavy and top-usage bands widen across cohorts. Value: the heavy line drops from \$121 to \$52 before recovering to \$70, while one-shot, weak week-1 and habit-light stay flat — which is what makes this a heavy-user story rather than a general decline.

Heavy users are arriving in far greater numbers and each is worth less. Two explanations, opposite fixes: they are self-selecting onto cheaper plans (packaging), or wider acquisition reaches a less committed audience (channel). The \$70-vs-\$121 gap is about one plan tier, which favours packaging — **plan mix by cohort settles it.**

Retention is drifting too: first renewal 58% → 48% (±1–3pp).

first-renewal rate by cohort (monthly, 95% CI)



April stands apart; May, June and July sit inside each other's confidence intervals, and August is the lowest of the five.

→ the \$117 LTV is a backward-looking average. On August's traffic it would come out lower, and that is the number to use for CAC decisions today.

7. So what?

finding	action
49% churn at the first bill, 19–31% after; leak proportional across segments	one problem, one target: the first renewal. Not a segment-specific programme
36% of cycle-1 revenue comes from accounts generating nothing	ship the dormant-payer list as a trigger — it is next cycle's churn, and today's refund risk
expansion never lifts NRR past 100%	upsell cannot substitute for fixing the first bill
LTV spans 4.6× across segments	CAC ceilings per segment, not one blended target
cost ceiling runs backwards to LTV (\$0.64 → \$0.04)	get cost per generation before scaling acquisition of heavy users
heavy share up 3×, heavy ARPU down 42%	check plan mix by cohort; if packaging, reprice the heavy tier
tail churn moves LTV 1.8×	quote LTV as a range

What would change these numbers most, in order: cost per image and video generation; CAC by channel; one more quarter of billing data, which would replace the tail assumption with measurement and roughly halve the range.